

Cloud Security Essentials TASK -5

- 1. Set up a free trial account on any major cloud provider (AWS, Azure, or Google Cloud) and deploy a basic virtual machine (VM) using default security settings.**

Ans :

- Create a free AWS account by visiting the AWS website.
- Complete account registration using your email address, password, contact details, and payment information.
- Log in to the AWS Management Console.
- From the AWS Console, search for and open EC2 (Elastic Compute Cloud).
- Select a suitable AWS Region (for example, Asia Pacific (Mumbai) – ap-south-1).
- Click Launch Instance.
- Enter a name for the virtual machine:
 - My-Test-VM
- Choose an operating system image (AMI), such as:
 - Amazon Linux 2023
 - Ubuntu Server
- Select a free-tier eligible instance type
- t2.micro / t3.micro
- Create or select a key pair for secure login:
 - Click Create new key pair.

- Enter a key name (e.g., my-vm-key).
 - Download the .pem file.
 - Configure network settings using the default security settings:
 - Default VPC
 - Default subnet
 - Enable SSH access (port 22) for Linux VM
 - Allow HTTP access only if required
 - Keep the default storage settings.
 - Review all VM configuration details.
 - Click Launch Instance.
- 2. Use an open-source vulnerability scanner like OpenVAS or Nikto to scan your deployed cloud VM for security vulnerabilities and save the scan report.**

Ans :

- Launch your deployed cloud VM (AWS EC2 instance).
- Note down the VM's public IP address.
- Install OpenVAS/Greenbone Vulnerability Scanner on a separate scanning machine or security testing VM.
- Update the OpenVAS vulnerability database
 - `sudo greenbone-feed-sync`
 - Start the OpenVAS services
 - `sudo gvm-start`
 - Open the OpenVAS web interface in a browser
 - `https://<scanner-IP>:9392`
- Log in using your OpenVAS credentials.

- Navigate to
- Scans → Tasks → New Task
- Create a new scan task.
- Enter the target cloud VM IP address

3. Identify at least three vulnerabilities from your scan report, and for each, document the vulnerability, its potential impact, and a recommended mitigation strategy in a table format.
Hint: For mitigation, consider changes to firewall rules, patching, or disabling unused services.

Ans :

Vulnerability Identified	Potential Impact	Recommended Mitigation Strategy
Open SSH Port (Port 22 Exposed to Internet)	Attackers may perform brute-force attacks to gain unauthorized access to the virtual machine.	Restrict SSH access using firewall/security group rules. Allow access only from trusted IP addresses, use SSH keys instead of passwords, and enable MFA where possible.
Missing Security Updates / Outdated Software Packages	Attackers can exploit known vulnerabilities in outdated operating systems or applications to	Regularly update and patch the operating system and installed applications. Enable automatic security updates and maintain a

Vulnerability Identified	Potential Impact	Recommended Mitigation Strategy
	compromise the VM.	patch management schedule.
Unnecessary Open Services and Ports	Exposed services may provide entry points for attackers and increase the attack surface of the cloud VM.	Disable unused services, close unnecessary ports in firewall/security group settings, and allow only required network traffic.
Weak Default Configuration	Default settings may reveal system information or allow unauthorized access.	Harden the VM configuration, remove default accounts, apply security best practices, and review permissions regularly.
Missing Security Headers (if a web server is running)	Attackers may exploit browser-based attacks such as clickjacking or content injection.	Configure proper HTTP security headers such as Content-Security-Policy, X-Frame-Options, and X-Content-Type-Options.

4. Pick one vulnerability from your findings and implement the recommended mitigation (for example, update a package, close an open port, or adjust a firewall rule), then rerun your vulnerability scan to verify if the issue is resolved.

Ans :

Scan Result	Before Mitigation	After Mitigation
Vulnerability	SSH Port 22 open to all users	SSH access restricted
Risk Level	Medium/High	Reduced
Firewall Rule	Port 22 open to 0.0.0.0/0	Port 22 allowed only from trusted IP
Scan Status	Vulnerability detected	Vulnerability resolved

5. Share your vulnerability scan results and mitigation steps with your group using a shared document or chat, and discuss how these issues might appear in popular apps like Zomato or Paytm if left unaddressed.

Ans :

Vulnerability	Risk/Impact	Mitigation Steps Implemented
Open SSH Port (Port 22 exposed publicly)	Attackers could perform brute-force attacks and attempt unauthorized access to the cloud VM.	Restricted SSH access by updating firewall/security group rules to allow only trusted IP addresses.
Missing Security Updates	Attackers could exploit known vulnerabilities in outdated software components.	Updated system packages and applied the latest security patches.

Vulnerability	Risk/Impact	Mitigation Steps Implemented
Unnecessary Open Services	Extra exposed services increase the attack surface and provide possible entry points for attackers.	Disabled unused services and closed unnecessary network ports.

Steps to Share Results with the Group

1. Create a shared document using:
 - Google Docs
 - Microsoft OneDrive
 - Any team collaboration platform
2. Add the following details:
 - Cloud platform used (AWS/Azure/GCP)
 - VM configuration
 - Vulnerability scan tool used (OpenVAS/Nikto)
 - Detected vulnerabilities
 - Risk impact
 - Mitigation actions performed
 - Before-and-after scan results
3. Share the document link with group members.
4. Discuss the findings with the team and collect feedback